



Detecting the Climate Signal in Ocean Outlet Glaciers

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Introduction

Mass loss from the Greenland and Antarctic ice sheets contributes significantly to global sea level rise.

• Since 1992, Greenland has lost about 170 billion tons of ice per year, and Antarctica about 90 billion tons per year.⁴

Glacier flow dynamics play a key role in ice mass loss

Along with a warming climate, there is also significant year over year Temperature variability the order of 1-2 degrees Celsius, and precipitation variability of order 1 mm/day⁵.

• However, glacier flow responds to climate change over years to centuries, causing complex and delayed ice sheet reactions.

Distinguishing climate change signals from natural variability in glacier flow remains challenging.

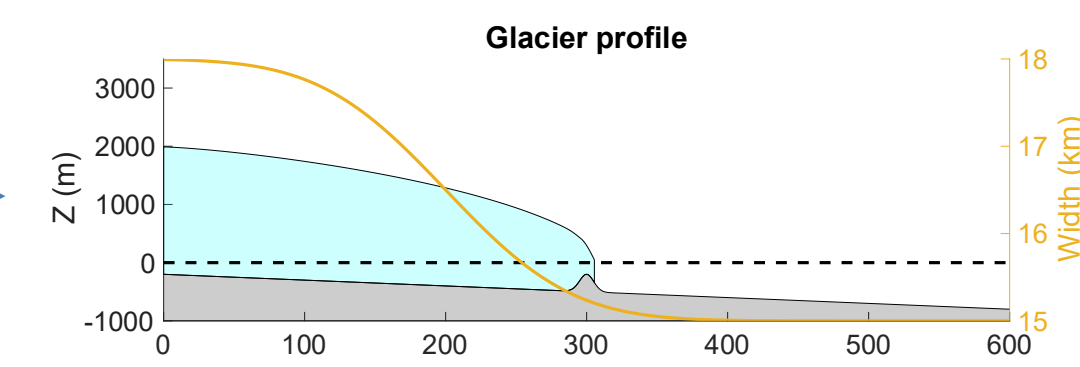
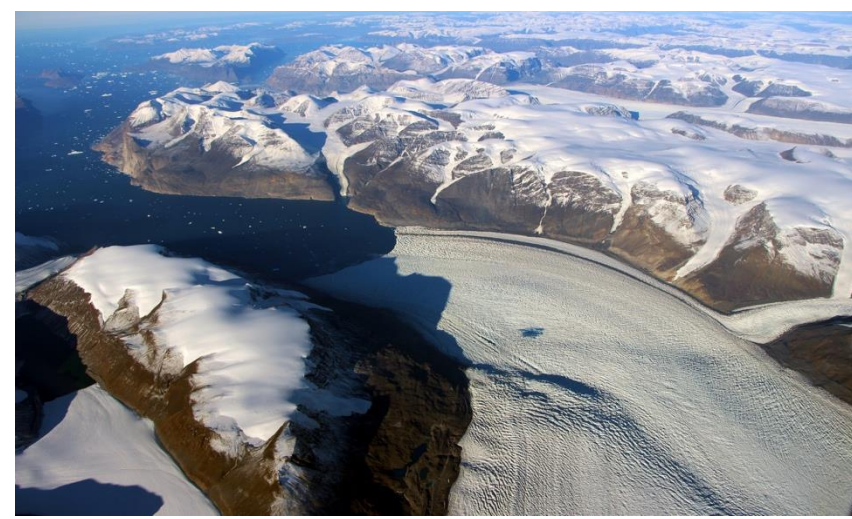
• Identifying when and where these dynamic signals emerge is essential for accurate sea level rise predictions.

Research Question

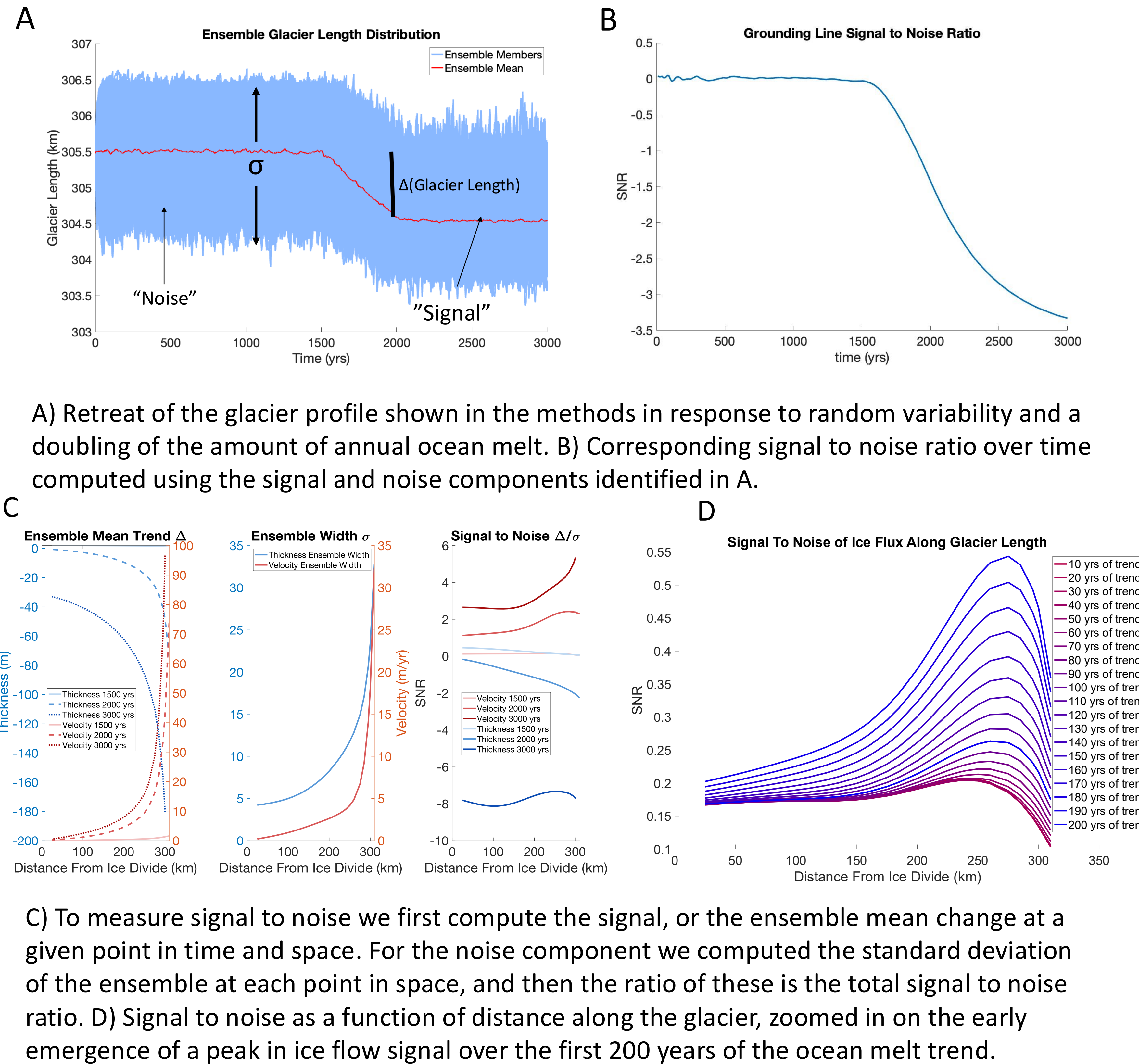
Where and when will the climate driven changes in ice flow emerge in glaciers?

Methods

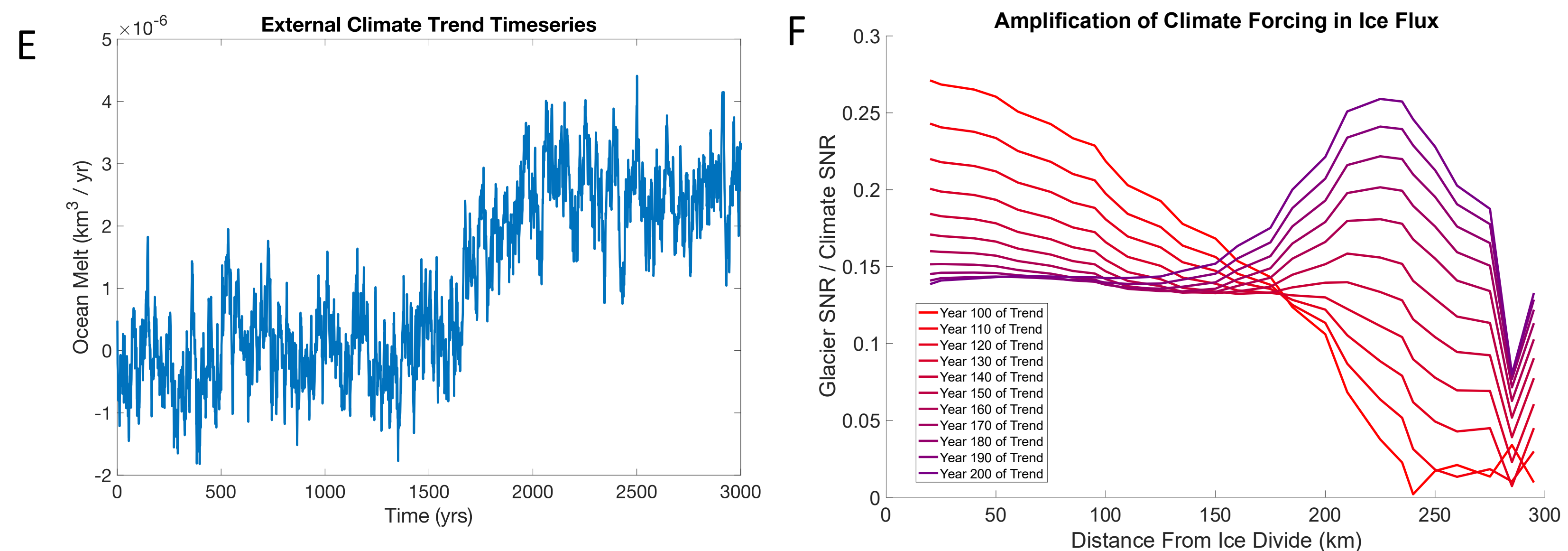
- We use a numerical model that simulates ice flow and basal sliding in one dimension, simplifying a glacier into its most basic attributes.^{1,2}



- Run 1000 simulations which all start from the same initial glacier steady state
- Apply a unique timeseries of random climate variability to each identical glacier allow for generation of internal variability between ensemble members.
- After year 1500, we apply the same linear trend in ocean induced melting at the front of the glacier.
- Signal ($\Delta(\text{glacier variable})$):** Estimate linear trends at fixed points along the glacier across all the glaciers for each glacier variable, such as ice flow velocity, ice flux through a region, or ice thickness.
- Noise (σ):** Take the standard deviation of quantities of interest across tall of the glaciers in the ensemble at the end of the 1500-year period with only random climate variability.
- Signal to Noise:** take the ratio of the two:
$$SNR_{thick} = \frac{\Delta(\text{glacier variable})}{\sigma}$$
- Computing the signal to noise ratio (SNR) of the forcing to that of the glacier's response, and assess how the glacier amplifies³ or filters climate changes: $Amp = \frac{Glacier\ SNR}{Climate\ SNR}$



C) To measure signal to noise we first compute the signal, or the ensemble mean change at a given point in time and space. For the noise component we computed the standard deviation of the ensemble at each point in space, and then the ratio of these is the total signal to noise ratio. D) Signal to noise as a function of distance along the glacier, zoomed in on the early emergence of a peak in ice flow signal over the first 200 years of the ocean melt trend.



E) Finally, extracting the climate time series signal and noise components and computing its SNR. F) Compute the ratio of the two to estimate the level of climate filtering or amplification that a particular glacier applies to the external forcing.

Conclusions

- Ensemble simulations allowed us to more effectively statistically characterize natural variability and trends
- By analyzing each point in time and along the glacier, we can observe how a trend emerges from background variability both spatially and in time. In many cases, a distinct peak in signal to noise emerges over time, particularly in velocity and then ice flux. This is persistent across multiple geometries.
- Amplification displays a more complex spatial pattern. but for this glacier with a steep bump, Values below 1 suggest that the climate trend is being filtered by the glacier.
- Moving forward, we aim to expand the scope of our simulations to capture a wider variety of glacial settings and continue refining climate forcing scenarios to better assess regional vulnerability to climate change.
- Ultimately, this analysis helps to visualize and anticipate how and when a glacier will respond to a changing climate that also has natural variability, and where we might look in order to observe the most significant changes.

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